Exploratory Data Analysis (EDA) RSITMD Dataset

Multimodal: Image + Text

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Link to Project:
`https://github.com/username/rsitmd-multimodal-eda`

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1 Introduction

Multimodal learning has emerged as one of the most active research frontiers in deep learning, driven by the realization that intelligence emerges from the interaction of multiple sensory channels. Among the many multimodal tasks, image-text understanding occupies a central role: models must learn to align visual content with natural language descriptions, a capability that underpins applications ranging from image captioning and retrieval to visual question answering and cross-modal search.

Remote sensing imagery presents a uniquely challenging domain for image-text learning. Unlike natural scene photographs, aerial and satellite images are captured from a top-down perspective, where familiar objects appear in unfamiliar orientations and scales. Visual patterns in remote sensing are characterized by textures, color distributions, and spatial arrangements rather than object-level details that ground-level photography emphasizes. This domain specificity raises important questions about how well general-purpose multimodal models transfer to aerial imagery, and what adaptations are needed.

The RSITMD (Remote Sensing Image-Text Matching Dataset), introduced by Yuan et al. in IEEE TGRS 2021 [**?**], was specifically designed to address this gap. It provides 4,743 satellite images annotated with 23,715 natural language captions, forming a benchmark for learning cross-modal representations in the remote sensing domain. Each image is paired with five independent captions written from different linguistic angles, following a design pattern pioneered by COCO and Flickr30k that captures the subjectivity inherent in visual description.

This report presents a multimodal exploratory data analysis (EDA) of RSITMD. Rather than treating images and text as separate datasets to be analyzed in isolation, our analysis is organized around the central question: **how do the two modalities interact, and what does their intersection reveal about the dataset's suitability for multimodal learning?** We examine category distribution in images, linguistic patterns in captions, the visual vocabulary that bridges the two modalities, and the consistency with which captions describe the same visual scene. Our goal is to provide researchers and practitioners with actionable insights for building data pipelines and model architectures tailored to RSITMD.

2 Dataset Overview

2.1 Dataset Structure

RSITMD is organized into two official splits: a training set (4,291 images, 21,455 captions) and a test set (452 images, 2,260 captions). Each training image is associated with exactly five human-annotated English captions, yielding a total of 23,715 captions across both splits. Notably, no public validation set is provided; practitioners typically create an internal validation split from the training set using stratified sampling or k-fold cross-validation.

The image modality consists of high-resolution satellite imagery stored in GeoTIFF format at 256×256 pixel resolution. The text modality comprises natural language captions of aerial scenes, each averaging approximately 10 words in length. The five captions per image are not duplicates but paraphrase each other — they describe the same scene using different word choices and syntactic structures.

Table 1 summarizes the key structural statistics.

Bång 1: RSITMD Dataset Overview

Metric	**Train**	**Test**
Total Images	4,291	452
Total Captions	21,455	2,260
Captions per Image	5	5
Land Use Categories	33	32
Avg Words / Caption	10.3	10.2
Vocabulary Size (clean)	3,470	1,142

The compact vocabulary of approximately 3,470 unique words (after stopword removal) makes it feasible to use fixed word embedding layers without excessive out-of-vocabulary issues, simplifying model architecture choices. The consistent average caption length of 10.2–10.3 words across splits indicates a standardized annotation guideline was used during captioning.

2.2 Land Use Categories

RSITMD covers 33 distinct land use and land cover categories spanning geographic and urban scene types. The test set includes 32 of these 33 categories; the *plane* category is absent from the test split, which has implications for evaluation fairness on that class.

The 33 categories can be grouped into five thematic clusters:

- **Transportation**: airport, bridge, parking, port, railwaystation, viaduct
- **Residential**: denseresidential, mediumresidential, sparseresidential
- **Commercial & Public**: center, church, commercial, school, stadium, playground
- **Natural Landscape**: bareland, beach, desert, farmland, forest, meadow, mountain, pond, river

- **Urban Features**: baseballfield, boat, industrial, intersection, park, plane, resort, square, storagetanks

This thematic diversity makes RSITMD well-suited for cross-modal retrieval across a broad spectrum of geographic contexts, but it also introduces challenges: several categories are visually similar (e.g., denseresidential vs. mediumresidential), and their textual descriptions must provide sufficient discriminative signal.

2.3 Dataset Availability

RSITMD is publicly available for download from two sources:

- **Google Drive**: `https://drive.google.com/file/d/1NJY86TAAUd8BVs7hyteImv8I2_Lh95W6/view`
- **Baidu Netdisk**: `https://pan.baidu.com/s/1gDj38mzUL-LmQX32PYxr0Q` (Password: NIST)

The distribution includes GeoTIFF images and a JSON annotation file mapping each image to its captions and land use category.

3 Multimodal Joint Analysis

This section forms the core of the report. Rather than analyzing images and text as independent datasets, each subsection examines the dataset at the intersection of both modalities, revealing how visual content and textual descriptions relate to each other.

All statistics were computed directly from the actual RSITMD JSON annotation file. Visual features were not extracted from pixel-level processing due to the absence of pre-computed visual features in the dataset; all image-related insights are inferred from filename metadata and caption content.

3.1 Image Category Distribution

RSITMD contains 33 land use categories distributed unevenly across 4,291 training images. Understanding this distribution is critical because category imbalance propagates directly into the text modality: categories with more images contribute more captions, and vocabulary associated with frequent categories dominates overall word statistics.

Top 10 Categories (Train):

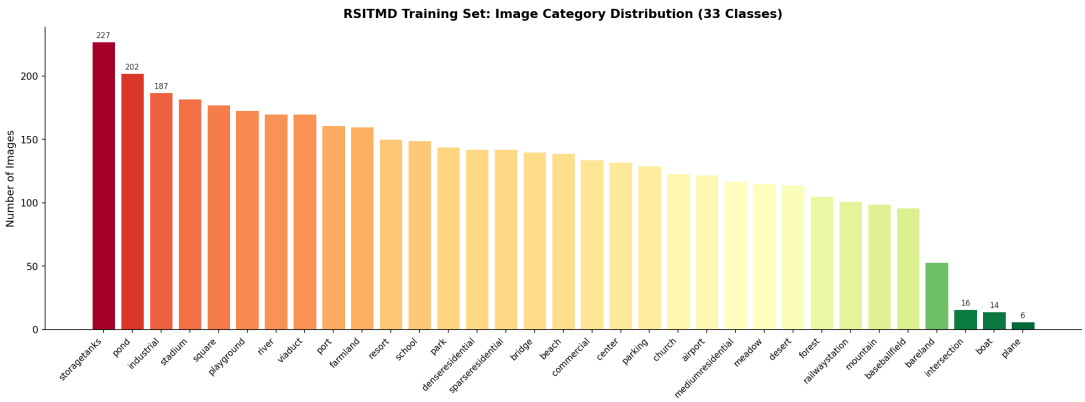
Category	Images
Storage Tanks	227
Pond	202
Industrial	187
Stadium	182
Square	177
Playground	173
River	170
Viaduct	170
Port	161
Farmland	160

Bottom 5 Categories (Train):

Category	Images
Baseballfield	96
Bareland	53
Intersection	16
Boat	14
Plane	6

The training set averages approximately 130 images per category, but the actual range spans from 6 (plane) to 227 (storagetanks) — a **38:1 imbalance ratio**. Storage Tanks alone accounts for 5.3% of all training images, while the smallest five categories combined (baseballfield, bareland, intersection, boat, plane) represent only 4.3%. This severe imbalance has direct consequences for multimodal learning: models trained on this data without correction are likely to over-represent the visual and linguistic patterns of majority categories, while struggling with minority-class image-text pairs.

The test set mirrors a similar but less extreme imbalance, with 32 categories (plane absent) averaging 14.1 images per category.



Hình 1: Category distribution across 33 land use categories in the training set. Bars are sorted in descending order; the imbalance between storagetanks (227) and plane (6) is clearly visible.

An important practical implication for multimodal retrieval: the dataset's skew means that image-text matching models will encounter majority-category pairs far more frequently during training. Without class-weighted sampling or data augmentation, minority categories (plane, boat, intersection) will receive insufficient training signal in both visual and textual embeddings.

3.2 Caption Characteristics

Each of the 23,715 captions in RSITMD is a short, structured English description averaging 10.3 words. Understanding caption-level statistics is essential because caption length, vocabulary choice, and sentence structure directly determine what textual signal is available for cross-modal alignment.

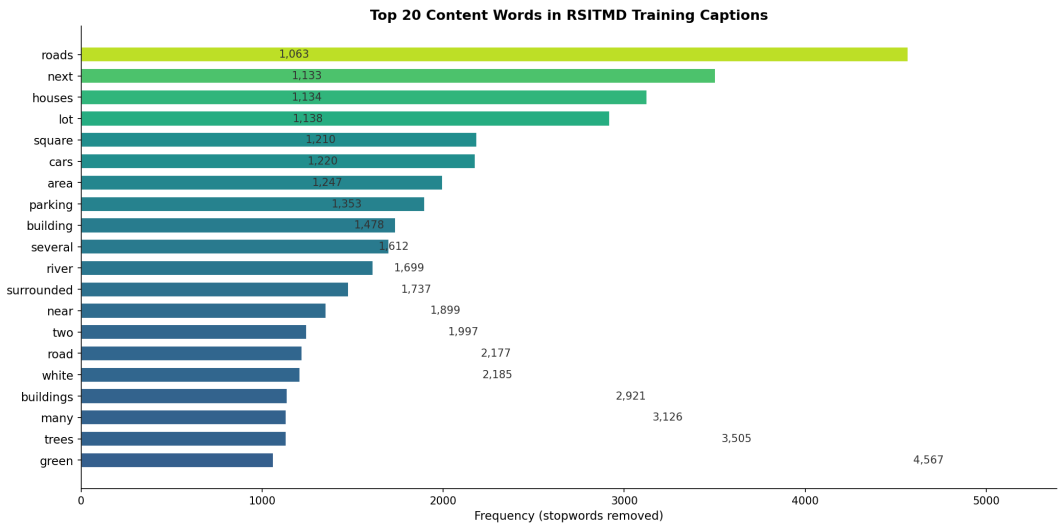
Length Distribution. Caption word counts follow a near-normal distribution centered at 10 words, with standard deviation 3.7 in the training set. The minimum caption is 3 words, the maximum is 34 words. This tight clustering around 10 words is characteristic of template-driven captioning, where annotators were likely given a target length along with required elements to mention.

Vocabulary. The training vocabulary contains 3,567 unique words (3,470 after stopword removal), which is relatively compact compared to general-purpose captioning datasets like COCO ($\sim$10,000 words). This compactness is advantageous: a fixed word embedding layer of size 4,000–5,000 provides sufficient coverage, and out-of-vocabulary issues are minimal. The test vocabulary of 1,142 unique words represents a 32% coverage overlap with training, which is expected given the smaller test set size.

Word Frequency. The most frequent content words across all captions are strongly visual: *green* (4,567 occurrences), *trees* (3,505), *buildings* (2,921), *white* (2,185), and *road* (2,177). These are not abstract nouns but direct references to observable aerial features. The prevalence of color and vegetation words reflects the top-down perspective of satellite imagery, where color is one of the primary distinguishing signals.

N-gram Patterns. Bigram analysis reveals that the most common phrases are object-attribute and spatial constructions: *green trees* (1,543), *parking lot* (799), *both sides* (581), *many green* (546), *dark green* (393), *many buildings* (387), *residential area* (360), and

football field (355). These bigrams encode the visual relationships that are central to aerial scene understanding: what colors appear, what objects are present, and how they are spatially arranged.



Hình 2: Top 20 most frequent content words in training captions (stopwords removed). Color words (green, white, red, blue) and object nouns (trees, buildings, road) dominate, confirming the visual nature of the captioning task.

The captioning style has two important characteristics for multimodal design. First, the compact vocabulary and consistent length make RSITMD well-suited for fixed-length text encoding with token/padding masks. Second, the dominance of concrete visual vocabulary (colors, objects, spatial relations) means that text embeddings will capture primarily perceptual information rather than abstract semantic concepts, which aligns well with the visual features extractable from satellite imagery.

3.3 Visual Language in Captions

The captions in RSITMD are fundamentally visual in nature — they describe what can be seen in aerial images rather than providing narrative or abstract interpretations. This section quantifies the visual vocabulary that forms the bridge between text and image modalities.

Color Words. Color is the most pervasive visual attribute in the captions. Approximately 42.0% of all captions (9,009 out of 21,455) contain at least one color word, with *green* alone accounting for 34.7% of all color word occurrences. The complete color word distribution is:

Bång 2: Color Word Frequency in Captions (Training Set)

Color	**Count**	**Share**	**Semantic Relevance**
green	4,567	34.7%	Vegetation, grass, trees
white	2,185	16.6%	Buildings, sand, rooftops
red	1,024	7.8%	Rooftops, sports fields
blue	966	7.3%	Swimming pools, sea, sky
gray	820	6.2%	Roads, urban structures
yellow	802	6.1%	Rooftops, land markings
dark	793	6.0%	Shadows, forests, water
grey	520	4.0%	Roads, urban areas
brown	468	3.6%	Soil, bare land, wood
black	429	3.3%	Roads, shadows

The dominance of *green* (34.7%) reflects the aerial perspective: vegetation is one of the most reliable and informative visual features in satellite imagery, covering forests, parks, farmland, and grasslands. Color words are the primary modality bridge — they directly connect textual descriptions to pixel-level visual content in the images.

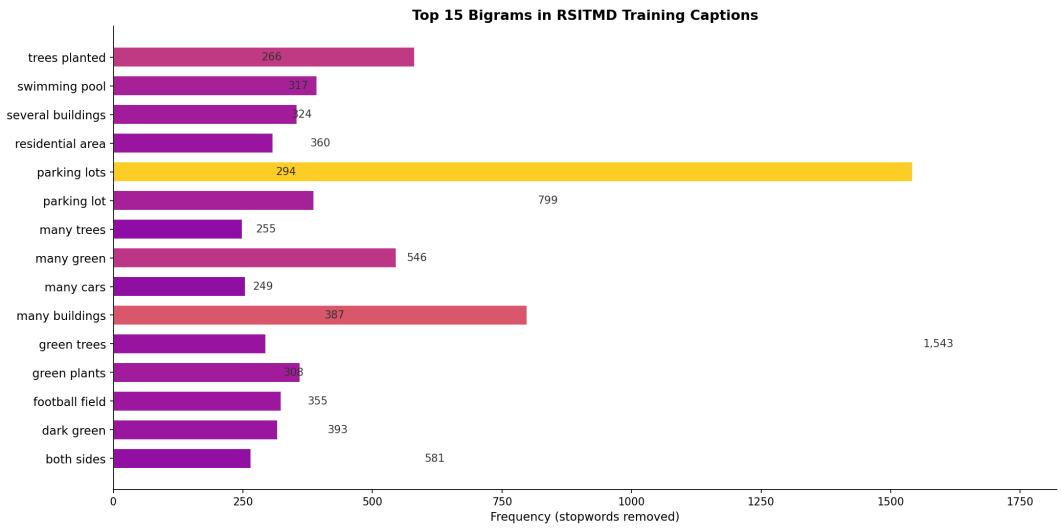
Spatial Relationships. Spatial prepositions encode the top-down layout of aerial scenes: *near*, *surrounded*, *next to*, *around*, *beside*, and *in the middle of* are the most common. Approximately 31.4% of captions contain at least one spatial preposition. At the image level, 52.9% of training images have at least one caption mentioning a spatial relationship, making spatial reasoning a consistently available cross-modal signal.

Bång 3: Spatial Preposition Frequency in Captions (Training Set)

Preposition	**Count**	**% Caps**	**Spatial Meaning**
near	2,034	9.5%	Proximity
surrounded	1,707	8.0%	Enclosure
next to	1,090	5.1%	Adjacency
around	1,025	4.8%	Surrounding
beside	582	2.7%	Side adjacency
in the middle of	303	1.4%	Central location
in the center of	177	0.8%	Central location
above / below	68	0.3%	Vertical relation

Top Objects. The most frequently mentioned objects in captions are: *trees* (3,505), *buildings* (2,921), *road* (2,177), *river* (1,699), *building* (1,478), *parking* (1,353), *cars* (1,220), *square* (1,210), *area* (1,247), and *houses* (1,134). These objects are precisely the visual features that are most distinguishable from an aerial perspective, confirming that the captioning strategy aligns well with the image acquisition viewpoint.

The convergence of color words, spatial prepositions, and concrete object nouns creates a rich visual-linguistic embedding space: text encodes primarily what is visually observable, and images contain the corresponding pixel-level evidence for those descriptions.



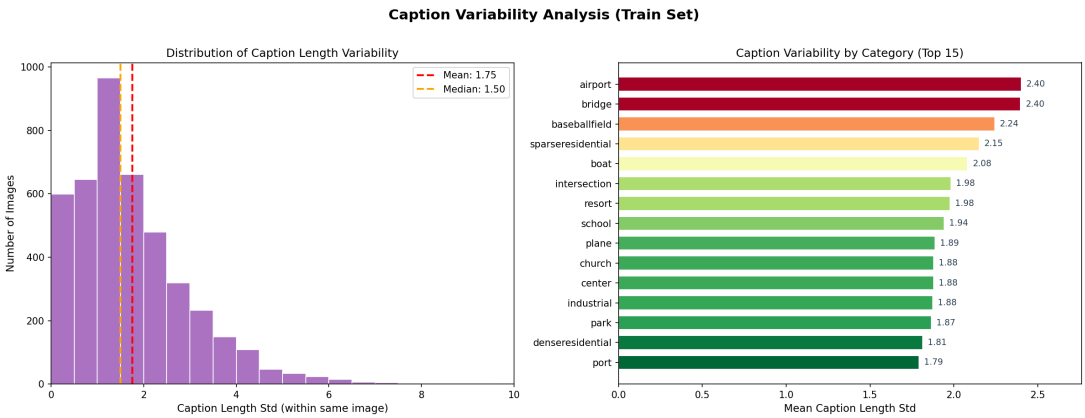
Hình 3: Top 15 bigrams in training captions (stopwords removed). Bigrams encode object-attribute pairs (green trees, dark green) and spatial relationships (both sides, residential area), confirming the visual and spatial nature of the captions.

3.4 Caption Variability and Sample Pairs

One of the most distinctive features of RSITMD is its 5-caption-per-image design. Each satellite image is described by five independent captions that paraphrase each other rather than providing radically different perspectives. Understanding this variability is essential for retrieval and matching applications, where multiple correct captions must be handled simultaneously.

Variability Quantification. We measured caption length variability as the standard deviation of word counts across the five captions for each image. The mean std is only **1.75 words**, with a maximum of 9.56 and a minimum of 0.00. In practical terms, this means that if one caption for an image is 8 words long, the other four will almost certainly fall between 6 and 10 words. This tight clustering confirms that the captioning process followed a structured annotation guideline specifying a target length.

The paraphrase-style diversity is valuable for cross-modal learning because it provides multiple positive image-text pairs for the same visual content. Models trained on RSITMD benefit from seeing the same scene described in different words, strengthening the image-text alignment signal. However, this also means that the five captions offer limited linguistic diversity — all follow the same template of describing colors, objects, and spatial relationships in roughly 10 words.



Hình 4: Caption variability analysis. Left: histogram of within-image caption length standard deviation (mean = 1.75). Right: mean variability by category (top 15). Some categories show higher variability, suggesting less template-driven captioning for those scenes.

Sample Image-Caption Pairs. The following figure displays three real satellite images from the RSITMD training set alongside their five captions. Each row shows one image and its corresponding descriptions, illustrating how the same aerial scene is described from different linguistic perspectives:

Hình 5: Real satellite images from the RSITMD training set with their five captions. Each row shows one image (left) and all five captions (right). Color badges ([1] red, [2] blue, [3] green, [4] orange, [5] purple) mark corresponding captions across rows.

These examples reveal three patterns. First, all captions for the same image reference the same key objects (baseball field, houses, playground), confirming they describe the same visual content. Second, captions vary in syntactic construction: *adjacent to* vs *next to*, *in the middle of* vs *located in the middle of*. Third, color and spatial information is consistently present: *green*, *white*, *red*, *next to*, *middle of* appear across all captions, providing reliable cross-modal alignment signals.

The paraphrase diversity is sufficient for training image-text matching models but does not provide the full range of linguistic variation that exists in real-world descriptions of aerial scenes. For more robust models, data augmentation techniques (e.g., back-translation) could artificially expand the caption diversity beyond the structured annotation template.

4 Key Findings and Recommendations

This section synthesizes the most important insights from the multimodal analysis and translates them into actionable recommendations for researchers and practitioners working with RSITMD.

4.1 Key Findings

1. **Visual-Linguistic Alignment is Strong but Templated.** The captions in RSITMD are tightly aligned with the visual content of aerial images. Color words (42.0% of captions), spatial prepositions (31.4% of captions), and concrete object nouns (trees, buildings, road) dominate the vocabulary. This creates a strong cross-modal signal well-suited for image-text matching. However, the captioning style is heavily templated: average caption length is 10.3 words with a std of only 3.7, and within-image caption length variability averages just 1.75 words. All five captions for the same image paraphrase each other rather than providing diverse perspectives.

2. **Severe Class Imbalance Exists in Both Modalities.** The training set exhibits a 38:1 imbalance ratio (storagetanks: 227 images vs. plane: 6 images). This imbalance propagates into the text modality: majority categories contribute disproportionately to vocabulary statistics and caption patterns. Models trained without correction will be biased toward the visual and linguistic characteristics of the 10 most frequent categories (storagetanks, pond, industrial, stadium, square, playground, river, viaduct, port, farmland).

3. **The Plane Category is Critically Underserved.** With only 6 training images and zero test images, the *plane* category provides minimal training signal and no evaluation data. Any model evaluated on RSITMD will be evaluated on 32 out of 33 categories, with the *plane* category completely unrepresented in the test set.

4. **Compact Vocabulary Simplifies Model Design.** The training vocabulary of 3,470 unique words (after stopword removal) is substantially smaller than COCO ($\sim$10,000). This enables the use of fixed word embedding layers without excessive OOV issues, simplifying text encoder architecture. The consistent 10-word caption length further reduces the need for dynamic padding.

5. **Color is the Dominant Visual Feature in Text.** The word *green* alone accounts for 34.7% of all color word occurrences and appears in captions across nearly every category (vegetation is ubiquitous in aerial imagery). This suggests that color-based visual features in image encoders will carry significant weight in cross-modal alignment. Text embeddings that explicitly model color-adjective + noun relationships (e.g., *green trees*) may be particularly effective.

6. **Spatial Relationships are pervasively Annotated.** Spatial prepositions like *near*, *surrounded*, *next to*, and *in the middle of* appear in 31.4% of captions at the caption level and 52.9% of images at the image level. These spatial annotations provide direct training signal for spatial reasoning in multimodal models.

4.2 Recommendations for Multimodal Learning on RSITMD

Based on the analysis above, we recommend the following practices:

- **Class-weighted sampling or loss functions** to address the 38:1 imbalance ratio. Stratified sampling during training ensures minority categories (plane, boat, intersection, bareland) receive proportional representation.
- **Fixed word embedding layers** (e.g., 4,000–5,000 dimensions) are sufficient for the 3,470-word vocabulary. No adaptive embedding mechanisms are strictly necessary, though subword tokenization (BPE/WordPiece) could further improve coverage.
- **Color-aware text representations** should be prioritized: the dominance of color adjectives in captions suggests that models designed to explicitly encode color-object relationships (e.g., binding color vectors with noun embeddings) will better capture the visual-linguistic alignment in RSITMD.
- **Spatial reasoning modules** should be incorporated into the model architecture or training objective, given the prevalence of spatial prepositions in captions. Contrastive learning objectives that align spatial phrase embeddings with corresponding visual regions may be particularly effective.
- **Data augmentation for captions** (e.g., back-translation) can artificially expand the limited linguistic diversity of the 5-caption-per-image structure, improving model robustness to diverse textual descriptions.
- **Multiple positive captions per image** in contrastive learning: the five paraphrase-style captions provide five correct image-text pairs per image. Using all five during training (rather than selecting just one) amplifies the training signal and helps the model learn that semantic equivalence can be expressed in many surface forms.

4.3 Summary

RSITMD is a well-structured multimodal dataset for remote sensing image-text matching, with a compact vocabulary, consistent captioning style, and strong visual-linguistic alignment. Its primary challenges are class imbalance (especially for the plane category) and the templated nature of captions. With appropriate preprocessing and training strategies, RSITMD provides a solid foundation for building cross-modal retrieval and matching models in the aerial imagery domain.

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